

# ratecalc

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# 1 Introduction

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In the computation of plasma source terms  $S_{A_p,k}$  due to interactions between neutral atoms and charged particles[1], integrals like the following must be computed,

$$S_{A_p,k} = \Gamma \int d^3v f(\vec{v}) \langle \sigma_k(v_{rel}) \Delta A_p \rangle_p, \quad (1)$$

where  $\Gamma$  is the flux of neutral particles,  $f(\vec{v})$  is the neutral particle distribution function,  $p$  is a plasma species index,  $\sigma_k$  is the cross section for the  $k$ th reaction between the neutral and species  $p$ ,  $v_{rel}$  is the relative velocity of the two particles, and  $\Delta A_p$  is the change in a plasma quantity (e.g., density, momentum, energy, etc.) in a single collision.

By expressing  $\Delta A_p$  as a power series in the relative velocity, the  $S_{A_p,k}$  can be written as a linear sum of velocity moments of the distribution function weighted by  $\sigma_k$ . The primary purpose of this code is to compute these moments at user-specified parameters using cross section data accessed from external files.

Presently, the cross section data is assumed to reside in the ALADDIN database. The formats of the input file and common blocks reflect this assumption. The input information used to identify the data in the database are in the *ra* class. Output from this code is stored in netCDF files using the *xs* class (see file classes.web for descriptions of both classes).

If the requested data represent a fit, the parameters used to describe the fit and the name of the function required to evaluate it are retrieved from the database and stored in the output data file. Alternatively, if the data for a particular dependent variable are in a “table”, the input information specifies the range, number, and spacing of the corresponding independent variables. At each set of independent variables, the dependent variable (cross section, reaction rate, or higher moment) is evaluated and sent to the output file.

More specifically, there are now three types of tabular organizations:

**table\_calc** Is of the type described above. Cross sections are averaged over a distribution function to obtain reaction rates and higher moments.

**table\_eval** Is used to simply evaluate the ALADDIN fit and store the data (for, e.g., transferring cross section data or ALADDIN reaction rates).

**table\_external** Is used to handle extraneous (usually constants) data which cannot be obtained from ALADDIN or which might be peculiar to a specific reaction. Placeholders are generated in the netCDF files; the user can later insert manually the required data.

The integral in Eq. (1) is evaluated here in two different regimes. The first is applicable to situations in which a neutral atom (“test”) collides with plasma electrons (“background”). In this case, the velocity moments required are given by

$$\langle \sigma v v^n \rangle = \frac{4\pi}{\alpha m} \left( \frac{2}{m} \right)^{(n+2)/2} \int_0^\infty dE \, \sigma f(E) E^{(n+2)/2}, \quad (2)$$

where  $\sigma(E)$  is the cross section,  $f(E)$  is an arbitrary distribution function,  $m$  is the particle mass relating  $E$  and  $v$  via  $E = \frac{1}{2}mv^2$ , and  $\alpha$  is the normalization constant for the distribution function,

$$\alpha = \frac{4\pi}{m} \left( \frac{2}{m} \right)^{1/2} \int_0^\infty dE \, f(E) E^{1/2}.$$

In order for the numerical integrations to work effectively, it is necessary that  $E$  be in practical units; eV in this case.

More specifically, the assumptions made in arriving at Eq. (2) are:

1.  $\sigma$  is a function only of the magnitude of the relative velocities of the colliding particles.
2. The distribution function is a function only of the particle’s energy (or, equivalently, speed).
3. The velocity of the “test” particle is much less than the typical velocity of the “background” particles (i.e., defined by  $f$ ).

In the second regime, the velocities of the two colliding particles may be comparable. For example, this would be the case if both species were “heavy” (e.g., neither was an electron). The integration in Eq. (1) must then be done over the relative velocity. This can be done conveniently when the plasma species is described by a Maxwellian distribution function. It is conceivable that an arbitrary distribution could be specified in terms of orthogonal polynomials times a Maxwellian so that the required integral could be computed as a linear sum of integrals like those described here.

The velocity moments in this regime are expressed in terms of the standard integral[1]

$$I_{\ell,n} \equiv \frac{\exp(-u^2/u_p^2)}{\sqrt{\pi}uu_p^{n+1}} \int_0^\infty dv_{rel} v_{rel}^{2+n} \sigma_\ell(v_{rel}) \left\{ \exp\left[\frac{-v_{rel}(v_{rel}-2u)}{u_p^2}\right] - (-1)^n \exp\left[\frac{-v_{rel}(v_{rel}+2u)}{u_p^2}\right] \right\}, \quad (3)$$

where  $u$  is the velocity of the neutral particle, and  $u_p$  is the thermal velocity of the plasma particle. The latter is related to the plasma temperature by  $T_p = \frac{1}{2}m_1u_p^2$ ; the former is derived from the neutral energy in the same way. The subscript  $\ell$  is included to allow the treatment of elastic collisions ( $\ell \neq 0$ ). Note that Eq. (3) is not the same as  $\langle\sigma vv^n\rangle$  because of the need to treat angular dependence in the integrand of Eq. (1), e.g., for the momentum source rate. Quantities such as this can be found as a linear combination of the  $I_{\ell,n}$ . The results provided by this code is  $I_{\ell,n}u_p^n$ . Note that  $\ell = 0$  represents the “total” cross section for either inelastic or elastic collisions. In either case, its  $n = 0$  moment is simply referred to as the “reaction rate”, which is used in computing the local probability of the reaction. For elastic collisions,  $\ell = 1$  represents the diffusion cross section and is associated with momentum transport; its  $n = 0$  moment is designated “I\_1\_0” to distinguish it from the reaction rate. Higher moments are denoted by I\_0\_1\*up or I\_1\_1\*up ( $n = 0, \ell = 0$  or 1) and I\_0\_2\*up^2 or I\_1\_2\*up^2 ( $n = 1, \ell = 0$  or 1). There is also I\_2\_4\*up^4 ( $n = 4, \ell = 2$ ) that has thus far been used only outside of DEGAS 2 for the purpose of computing viscosity. Note that  $I_{\ell,n}u_p^n$  has the same dimensions as  $\langle\sigma vv^n\rangle$ . In practice, moments  $n > 0$  for  $\langle\sigma vv^n\rangle$  (i.e., for electron background) are not used.

In the expressions defining the neutral energy and plasma temperature, the mass  $m_1$  used is 1 amu. The independent variable is referred to as “specific energy” or “specific temperature” to indicate this. This allows the results to be easily rescaled for different isotopes. It is crucial to note the asymmetry between the neutral and plasma variables. The rates computed by this code are intended to be used primarily in Monte Carlo neutral transport codes. The objective of these codes is to, in some approximation, compute the neutral particle distribution function. Hence, this information is not known until the end of the calculation; in a Monte Carlo code, however, one will know the instantaneous *energy* of a test neutral particle. To carry out the kinetic calculation, the distribution of the background species must be known. This is usually assumed to be Maxwellian; the characteristic parameter in this case is the plasma *temperature*. So, it is usually the case that the independent variable “specific temperature” is to be associated with a plasma ion species, “specific energy” with a neutral species, and “temperature” with plasma electrons.

To facilitate numerical integration, the units used for the various velocities are  $\sqrt{\text{eV}/\text{amu}}$ ; neither  $u$  nor  $u_p$  should be  $\leq 0$ . However, the energy used as the argument for  $\sigma$  must have a specific mass. Its value (specified by *ra.mass* here) varies, depending on the origin of the cross section data; it is assumed to be in grams. The independent variable to be used with cross sections in these reactions should be “specific\_energy”. This will eliminate any confusion in subsequent uses of the data. An independent variable of “energy” should be used for electron impact processes, or other reactions in which the mass to use is clear.

For example, for the data described in Ref. [2] one would need to take *ra.mass* to be that of the charged reagent (electron or ion). On the other hand, the data in Ref. [3] use a mass of 1 amu for “heavy” particle reactions and the electron mass for electron impact processes.

```
"ratecalc.f" 1 ≡
@M FILE 'ratecalc.web'
```

The unnamed module. This subroutine just extracts the input filename from the argument line and passes it along to the reading subroutine.

```
"ratecalc.f" 1.1 ≡  
  program ratecalc  
    implicit_none_f77  
    implicit_none_f90  
    character*LINELEN arg  
    character*FILELEN filename  
  
    sy_decls  
  
    assert(arg_count( ) ≡ 1)  
    call command_arg(1, arg)  
    filename = arg  
    call read_input(filename)  
  
    stop  
end
```

⟨ Functions and subroutines 2 ⟩

## 2 Input for RATECALC

Read input data into common variables and set up defaults. The format of the input file is described in the examples. Note that some quantities are either determined locally or assumed to have specific values. Those variables are:

*ra\_num\** Set by counting the corresponding labels.

*xs\_num\_dep\_var* Set by counting the number of dependent variables specified.

*xs\_rank* Set by counting the number of independent variables.

- Note that when the data are a fit, the *xs\_table* array will effectively be of rank one, i.e., containing the fit coefficients.
- Also in this case *xs\_tab\_index* is set temporarily to the parameter used to specify the maximum number of fit coefficients since the user will not know this in general.

*xs\_units* Are assumed to be cgs or eV for temperature and energy.

*xs\_multiplier* Are then the corresponding MKS conversion factors.

⟨Functions and subroutines 2⟩ ≡

```

subroutine read_input(filename)
  implicit_none_f77
  ra_common
  xs_common
  implicit_none_f90

  character*FILELEN filename    // Input
  integer length    // Local
  character*LINELEN line

  st_decls
  xs_decls

  open (unit = disk_in, file = filename, form = 'formatted', status = 'old')

  if (¬read_string(disk_in, line, length)) then
    line = 'END'
  end if
  length = parse_string(line(: length))

loop1: continue

  if (line ≡ 'END')
    goto eof
  if (line ≡ 'DEPENDENT_VARIABLES') then
    call read_dependent(disk_in, line)
  else if (line ≡ 'INDEPENDENT_VARIABLES') then
    call read_indep(disk_in, line)
  else if (line ≡ 'REACTIONS') then
    call read_reaction_list(disk_in, line)
  end if

  goto loop1

eof: continue

  return
end

```

See also sections 2.1, 2.2, 2.3, 3, 3.1, 3.2, 4, 4.1, 4.2, 4.3, 4.4, 5, 5.1, 6, 6.1, 6.2, 6.3, and 6.4.

This code is used in section 1.1.

Read in data for dependent variables.

⟨Functions and subroutines 2⟩ +≡

```
subroutine read_dependent(unit, line)
  implicit none
  xs_common
  implicit none
  integer unit    // Input
  character*(*) line // Output
  integer length, p, b, e, i, j // Local

  st_decls
  xs_decls

  xs_num_dep_var = 0
```

next\_dep\_var: **continue**     // Label macros are defined in ratecalc.hweb

```
  assert(read_string(unit, line, length))
  assert(length ≤ len(line))
  length = parse_string(line(: length))
  if (line ≡ 'INDEPENDENT_VARIABLES')
    goto end_dep_var

  p = 0 // For each dep. var.:
  assert(next_token(line(: length), b, e, p)) // Read name
  xs_num_dep_var++
  assert(xs_num_dep_var ≤ xs_dep_var_max)
  xs_var0, xs_num_dep_var = read_text(line(b : e))

  assert(next_token(line(: length), b, e, p))
  xs_eval_name xs_num_dep_var = read_text(line(b : e)) // Eval. name

  if (next_token(line(: length), b, e, p)) then
    xs_spacing0, xs_num_dep_var = read_text(line(b : e)) // And spacing
  else
    xs_spacing0, xs_num_dep_var = 'unknown'
  end if

  goto next_dep_var
```

end\_dep\_var: **continue**

/\* Utilize cgs assumptions for units. Multipliers are conversions to MKS. \*/

```
do i = 1, xs_num_dep_var
  if (xs_var0, i ≡ 'cross_section') then
    xs_units0, i = 'cm2'
    xs_mult0, i = const(1., -4)
  else if (xs_var0, i ≡ 'reaction_rate') then // I0,0
    xs_units0, i = 'cm3/s'
    xs_mult0, i = const(1., -6)
  else if (xs_var0, i ≡ 'I_1_0' ∨ xs_var0, i ≡ 'I_1_0') then
    xs_units0, i = 'cm3/s'
    xs_mult0, i = const(1., -6)
  else if (xs_var0, i ≡ 'I_1_1*up' ∨ xs_var0, i ≡ 'I_0_1*up' ∨ xs_var0, i ≡ 'I_1_1*up') then
    xs_units0, i = 'cm4/s2'
    xs_mult0, i = const(1., -8)
  else if (xs_var0, i ≡ 'I_1_2*up2' ∨ xs_var0, i ≡ 'I_0_2*up2' ∨ xs_var0, i ≡ 'I_1_2*up2') then
```

```

      xs_units0, i = 'cm5/s3'
      xs_mult0, i = const(1., -10)
    else if (xs_var0, i ≡ 'I_2_4*up4') then
      xs_units0, i = 'cm7/s5'
      xs_mult0, i = const(1., -14)
    else if (xs_var0, i ≡ 'sigv_max') then
      xs_units0, i = 'cm3/s'
      xs_mult0, i = const(1., -6)
    else
      xs_units0, i = 'unknown'
  @#if 0
      xs_mult0, i = zero
  @#endif
      xs_mult0, i = one
    end if
  end do

  /* Set defaults for unused dependent variables */
  if (xs_num_dep_var < xs_dep_var_max) then
    do j = xs_num_dep_var + 1, xs_dep_var_max

      xs_rankj = 1
      xs_eval_namej = 'unknown'
      xs_var0, j = 'unknown'
      xs_spacing0, j = 'unknown'
      xs_units0, j = 'unknown'
      xs_mult0, j = zero
    do i = 1, xs_table_rank_max
      xs_tab_indexi, j = 1
      xs_vari, j = 'unknown'
      xs_spacingi, j = 'unknown'
      xs_unitsi, j = 'unknown'
      xs_multi, j = zero
      xs_mini, j = zero
      xs_maxi, j = zero
    end do
  end do
end if

return
end

```

Read in data for independent variables.

⟨Functions and subroutines 2⟩ +≡

```
subroutine read_indep(unit, line)
  implicit_none_f77
  xs_common
  implicit_none_f90
  integer unit    // Input
  character*(*) line // Output
@#if 0
  character*FILELEN tc // Test String
@#endif
  integer length, p, b, e, i, idep, j // Local
  integer dep_var xs_dep_var_max
  logical all_ints

  st_decls
  xs_decls

  do i = 1, xs_dep_var_max
    xs_rank_i = 0
  enddo
```

next\_line: **continue** // Labels are defined in ratecalc.hweb

```
  assert(read_string(unit, line, length))
  if (line ≡ 'REACTIONS')
    goto end_indep_var
  assert(length ≤ len(line))
  length = parse_string(line(: length))
```

p = 0

```
  assert(next_token(line(: length), b, e, p))
```

/\* We need to differentiate between the lines which list the indices of the dependent variables and those which enumerate the corresponding independent variables. The former should begin with a 1 or 2 digit integer; hence, the check on *xs\_dep\_var\_max*. Set *all\_ints* to true if these tokens are indeed integers. The independent variables are given by a multi-character string (*all\_ints* = false). One in current use does begin with a "1"; hence, the care exercised here. \*/

@#if 0

```
  tc = line(b : b)
```

```
  if (tc ≡ '0' ∨ tc ≡ '1' ∨ tc ≡ '2' ∨ tc ≡ '3' ∨ tc ≡ '4' ∨ tc ≡ '5' ∨ tc ≡ '6' ∨ tc ≡ '7' ∨ tc ≡
    '8' ∨ tc ≡ '9') then
```

@#endif

```
  if (b ≡ e) then
```

```
    all_ints = (lge(line(b : e), '1') ∧ lle(line(b : e), '9'))
```

```
  else if (e ≡ b + 1) then
```

```
    all_ints = (lge(line(b : e), '10') ∧ lle(line(b : e), '99'))
```

```
  else
```

```
    assert(xs_dep_var_max ≤ 99)
```

```
    all_ints = F
```

```
  end if
```

```
  if (all_ints) then
```

```
    idep = 1 // Read in list of dependent variables
```

```
    dep_var_1 = read_integer(line(b : e))
```

```

next_dep_var: continue
  if ( $\neg$ next_token(line(:length), b, e, p))
    goto next_line
  iddep++
  dep_var_iddep = read_integer(line(b:e))
  goto next_dep_var
else // And their corresponding independent variables
  xs_rank_dep_var1++
  assert(xs_rank_dep_var1 ≤ xs_table_rank_max)
  xs_var_xs_rank_dep_var1, dep_var1 = read_text(line(b:e))
  if (xs_var_xs_rank_dep_var1, dep_var1 ≡ 'none') then
    xs_rank_dep_var1 = 0 // Constant
  else // Its minimum
    if (next_token(line(:length), b, e, p)) then
      xs_min_xs_rank_dep_var1, dep_var1 = read_real(line(b:e))
    else
      xs_min_xs_rank_dep_var1, dep_var1 = zero
    end if // Maximum
    if (next_token(line(:length), b, e, p)) then
      xs_max_xs_rank_dep_var1, dep_var1 = read_real(line(b:e))
    else
      xs_max_xs_rank_dep_var1, dep_var1 = zero
    end if // Number of values
    if (next_token(line(:length), b, e, p)) then
      xs_tab_index_xs_rank_dep_var1, dep_var1 = read_integer(line(b:e))
    else
      if (xs_eval_name_dep_var1 ≡ 'fit' ∧ xs_rank_dep_var1 ≡ 1) then
        xs_tab_index_xs_rank_dep_var1, dep_var1 = ra_fit_coef_max
      else
        xs_tab_index_xs_rank_dep_var1, dep_var1 = 1
      end if
    end if // And spacing
    if (next_token(line(:length), b, e, p)) then
      xs_spacing_xs_rank_dep_var1, dep_var1 = read_text(line(b:e))
    else
      xs_spacing_xs_rank_dep_var1, dep_var1 = 'unknown'
    end if
  end if // Propagate to other
  if (iddep > 1) then // dependent vars.
    do i = 2, iddep
      xs_rank_dep_var_i = xs_rank_dep_var1
      if (xs_rank_dep_var1 > 0) then
        xs_var_xs_rank_dep_var1, dep_var_i = xs_var_xs_rank_dep_var1, dep_var1
        xs_min_xs_rank_dep_var1, dep_var_i = xs_min_xs_rank_dep_var1, dep_var1
        xs_max_xs_rank_dep_var1, dep_var_i = xs_max_xs_rank_dep_var1, dep_var1
        xs_tab_index_xs_rank_dep_var1, dep_var_i = xs_tab_index_xs_rank_dep_var1, dep_var1
        xs_spacing_xs_rank_dep_var1, dep_var_i = xs_spacing_xs_rank_dep_var1, dep_var1
      end if
    end do
  end if
end do

```

```

        end if
    end if
    goto next_line
end_indep_var: continue

    /* Assign cgs units with conversions to MKS */
do j = 1, xs_num_dep_var
    if (xs_rank_j > 0) then
        do i = 1, xs_rank_j
            if (xs_var_i,j ≡ 'energy' ∨ xs_var_i,j ≡ 'temperature') then
                xs_units_i,j = 'eV'
                xs_mult_i,j = electron_charge
            else if (xs_var_i,j ≡ 'specific_energy' ∨ xs_var_i,j ≡ 'specific_temperature') then
                xs_units_i,j = 'eV/amu'
                xs_mult_i,j = electron_charge / atomic_mass_unit
            else if (xs_var_i,j ≡ 'density') then
                xs_units_i,j = 'cm3'
                xs_mult_i,j = const(1., 6)
            else
                xs_units_i,j = 'unknown'
            @#if 0
                xs_mult_i,j = zero
            @#endif
                xs_mult_i,j = one
            end if
        end do
    end if
    /* Fill defaults for unused ranks */
    if (xs_rank_j < xs_table_rank_max) then
        do i = xs_rank_j + 1, xs_table_rank_max
            xs_tab_index_i,j = 1
            xs_var_i,j = 'unknown'
            xs_spacing_i,j = 'unknown'
            xs_units_i,j = 'unknown'
            xs_mult_i,j = zero
            xs_min_i,j = zero
            xs_max_i,j = zero
        end do
    end if
end do
return
end

```

Read in data for reactions. This also forms the starting point for the rest of the ratecalc code which fills in the tables and writes the netCDF files for each reaction.

⟨ Functions and subroutines 2 ⟩ +≡

```
subroutine read_reaction_list(unit, line)
  implicit_none_f77
  xs_common
  ra_common
  implicit_none_f90
  integer unit      // Input
  character*(*) line // Output
  integer length, p, b, e, i, idep // Local
  integer dep_var xs_dep_var_max
  logical and_section

  st_decls
  xs_decls

  /* The following declarations are required to write out reaction specification information (ra class)
  as a netCDF file.
      integer fileid           character*FILELEN fileout
  nc_decls           ra_ncdecl
```

```

*/
react_begin: continue    // Labels are defined in ratecalc.hweb

    if ( $\neg$ read_string(unit, line, length))
        line = 'END'
    if (line  $\equiv$  'END')
        goto eof
    assert(length  $\leq$  len(line))
    length = parse_string(line(: length))

    xs_name = line(: length)    // First line: reaction name

    assert(read_string(unit, line, length))    // Second line: mass and units
    length = parse_string(line(: length))
    p = 0
    assert(next_token(line, b, e, p))
    ra_mass = read_real(line(b : e))
    assert(next_token(line, b, e, p))
    ra_mass_units = read_text(line(b : e))

    assert(read_string(unit, line, length))    // Third line: data filename
    length = parse_string(line(: length))
    assert(length  $\leq$  len(line))
    ra_data_filename = line(: length)

    assert(read_string(unit, line, length))    // Fourth line: data version
    length = parse_string(line(: length))
    assert(length  $\leq$  len(line))
    xsection_version = line(: length)

    assert(read_string(unit, line, length))    // Fifth line: hier. labels
    length = parse_string(line(: length))
    assert(length  $\leq$  len(line))
    p = 0
    ra_num_hier = 0

next_hlab: continue

    if ( $\neg$ next_token(line, b, e, p))
        goto bool_sect
    ra_num_hier++
    ra_hier_labelra_num_hier = read_text(line(b : e))
    goto next_hlab

    /* Subsequent pairs of lines provide the Boolean search labels for this reaction. */
bool_sect: continue

    do i = 1, xs_num_dep_var
        ra_num_bool_andi = 0
        ra_num_bool_noti = 0
    end do

new_bool_sect: continue    // Start a new pair of lines

    /* If next line is blank or is the end of file, it indicates the end of this reaction specification. */
    idep = 0
    if ( $\neg$ read_string(unit, line, length))
        goto react_done    // End of file

```

```

    length = parse_string(line(: length))
    if (line(1:1) == '-')
        goto react_done

    p = 0
next_dep_var: continue    // First line: dependent variables

    if (~next_token(line, b, e, p))
        goto bool_list
    idep++
    dep_var(idep) = read_integer(line(b:e))
    assert(dep_var(idep) ≤ xs_num_dep_var)
    goto next_dep_var

bool_list: continue

    assert(read_string(unit, line, length))    // Second: corresponding labels
    length = parse_string(line(: length))
    p = 0
    and_section = T

next_bool: continue

    if (~next_token(line, b, e, p))
        goto new_bool_sect
    if (line(b:e) == '&!')
        and_section = F
    if (and_section) then
        do i = 1, idep
            ra_num_bool_and_dep_var_i++
            ra_bool_and_label_ra_num_bool_and_dep_var_i, dep_var_i = read_text(line(b:e))
        end do
    else
        do i = 1, idep
            ra_num_bool_not_dep_var_i++
            ra_bool_not_label_ra_num_bool_not_dep_var_i, dep_var_i = read_text(line(b:e))
        end do
    end if

    goto next_bool

    /* Reaction specification complete */
react_done: continue

    /* The ra class data can be written to a netCDF file if needed:
       fileout='test.nc'
       fileid=nccreate(fileout,NC_CLOBBER,nc_stat)          ra_ncdef(fileid)
       call ncendef(fileid,nc_stat)          ra_ncwrite(fileid)          call
       ncclose(fileid,nc_stat)
       Now fill tables and write data file */
       call fill_tables

       goto react_begin

eof: return
end

```

### 3 Fill Cross Section Tables

This routine carries out the loops over dependent and independent variables. Table values are obtained directly from ALADDIN (for fits) or via subroutine *table\_value*.

⟨ Functions and subroutines 2 ⟩ +=

```

subroutine fill_tables
  implicit_none_f77
  ra_common     // Commons and declarations
  xs_common
  rf_common
  implicit_none_f90
  integer fileid, j, k, l, m, b, e, p     // Local
  real zsigma, table_value
  real ind_var_minxs_table_rank_max, ind_var_deltaxs_table_rank_max, ind_varxs_table_rank_max
  character*FILELEN fileout
  character*LINELEN description

  xs_ncdecl
  nc_decls
  st_decls
  xs_decls
  ⟨ Memory allocation interface 0 ⟩
  ra_localcommon     // Local common
  xs_ragged_alloc(xs_data, xs_tab_index)

  do j = 1, xs_num_dep_var     // Main loop
    /* Get fit information from ALADDIN. */
    if (xs_eval_name_j ≠ 'table_external')
      call aladdin_read_coefs(ra_data_filename, ra_hier_label, ra_num_hier, ra_bool_and_label1, j,
        ra_num_bool_and_j, ra_bool_not_label1, j, ra_num_bool_not_j, fit_coef, num_fit_coef, level)

    if ((xs_eval_name_j ≡ 'table_eval') ∨ (xs_eval_name_j ≡ 'table_calc')) then
      /* For tables, find all delta values at this j */
      do k = 1, xs_table_rank_max
        call find_delta(xs_spacing_k, j, xs_min_k, j, one, xs_max_k, j, xs_tab_index_k, j, ind_var_min_k,
          ind_var_delta_k)
      end do

      /* The following three loops represent the present size of xs_table_rank_max; if it is increased
         or decreased, the number of loops here should be altered accordingly. */
      do k = 0, xs_tab_index3, j - 1
        ind_var3 = ind_var_min3 + areal(k) * ind_var_delta3
        if (xs_spacing3, j ≡ 'log')
          ind_var3 = exp(ind_var3)

        do l = 0, xs_tab_index2, j - 1
          ind_var2 = ind_var_min2 + areal(l) * ind_var_delta2
          if (xs_spacing2, j ≡ 'log')
            ind_var2 = exp(ind_var2)

          do m = 0, xs_tab_index1, j - 1
            ind_var1 = ind_var_min1 + areal(m) * ind_var_delta1
            if (xs_spacing1, j ≡ 'log')

```

```

        ind_var_1 = exp(ind_var_1)
        zsigma = table_value(j, ind_var)
        xs_data_table(m, l, k, j) = zsigma
    end do

    end do
end do     // End of “do” loops over xs_table indices

/* For fits, store name of evaluation function, number of fit coefficients, and the coefficients
   themselves. */
else if (xs_eval_namej ≡ 'fit') then
    xs_eval_namej = leval
    xs_tab_index1, j = num_fit_coef

    do m = 0, xs_tab_index1, j - 1
        xs_data_table(m, 0, 0, j) = fit_coefm+1
    end do

    /* For externally specified data, fill table with the appropriate number of zeroes. */
else if (xs_eval_namej ≡ 'table_external') then
    do k = 0, xs_tab_index3, j - 1
        do l = 0, xs_tab_index2, j - 1
            do m = 0, xs_tab_index1, j - 1
                xs_data_table(m, l, k, j) = zero
            end do
        end do
    end do
end do

else
    assert('Unknown_organization' ≡ ' ')
end if     // End for “if” on xs_eval_name
end do     // End of “do” over dependent variables

do j = xs_num_dep_var + 1, xs_dep_var_max
    do k = 0, xs_tab_index3, j - 1
        do l = 0, xs_tab_index2, j - 1
            do m = 0, xs_tab_index1, j - 1
                xs_data_table(m, l, k, j) = real_unused
            end do
        end do
    end do
end do

/* All done. Write and close netCDF file */
p = 0
assert(next_token(xs_name, b, e, p))

fileout = xs_name(b:e) || '.nc'
fileid = nccreate(fileout, NC_CLOBBER, nc_stat)

description = 'Reference_atomic_physics_data_for_a_reaction_in_degas2'
call ncattputc(fileid, NC_GLOBAL, 'description', NC_CHAR, string_length(description),
    description, nc_stat)

```

```
call ncattputc(fileid, NC_GLOBAL, 'data_version', NC_CHAR, string_length(xsection_version),  
              xsection_version, nc_stat)  
  
xs_ncdef(fileid)  
call ncendef(fileid, nc_stat)  
xs_ncwrite(fileid)  
  
var_free(xs_data_tab)  
  
call ncclose(fileid, nc_stat)  
  
return  
end
```

Table evaluation details. This routine uses the information about the dependent and independent variables to infer the type of evaluation which is required (cross section, electron reaction rate, ion reaction rate, etc.) and makes the appropriate function calls. The input parameters are  $j$  the index into the  $xs$  class variables for this dependent variable and  $ind\_var$  the values of the independent variables for this table entry.

⟨ Functions and subroutines 2 ⟩ +≡

```

function table_value( $j$ ,  $ind\_var$ )
  implicit_none_f77
  xs_common
  ra_common
  implicit_none_f90

  real table_value    // Function
  integer  $j$           // Input
  real  $ind\_var$   $xs\_table\_rank\_max$ 

  integer num_fit_coef_aladdin    // Local
  real norm_constant, infinite_integral, zsigma, zenergy, ztemperature
  double precision zte_aladdin, zdpsigma

  character*msg_length error

  external electron_rate_integrand,    /* External */
    electron_norm_integrand, ion_rate_integrand
  real electron_rate_integrand, electron_norm_integrand, ion_rate_integrand

  ra_localcommon
  xs_decls    /* For cross sections, presumably will only be evaluating fits to fill the table. Hence,
    moment_number is not needed. But, for higher moments, the variable moment_number must be
    set. Determine the required value from the contents of xs_var0, j. */
  if ( $xs\_var0, j \equiv$  'cross_section') then
    assert( $xs\_eval\_name_j \equiv$  'table_eval')
  else if ( $xs\_var0, j \equiv$  'reaction_rate') then
    moment_number = 0
  else if ( $xs\_var0, j \equiv$  'I_1_0'  $\vee$   $xs\_var0, j \equiv$  'I_1_0') then
    moment_number = 0
  else if ( $xs\_var0, j \equiv$  'I_1_1*up'  $\vee$   $xs\_var0, j \equiv$  'I_0_1*up'  $\vee$   $xs\_var0, j \equiv$  'I_1_1*up') then
    moment_number = 1
  else if ( $xs\_var0, j \equiv$  'I_1_2*up2'  $\vee$   $xs\_var0, j \equiv$  'I_0_2*up2'  $\vee$   $xs\_var0, j \equiv$  'I_1_2*up2') then
    moment_number = 2
  else if ( $xs\_var0, j \equiv$  'I_2_4*up4') then
    moment_number = 4
  @#if 0
  else if ( $xs\_var0, j \equiv$  'sigv_max') then
    assert( $xs\_eval\_name_j \equiv$  'table_external')
  else
    assert('Unknown_independent_variable'  $\equiv$  '_')
  @#endif
  end if

  /* Now split up procedures according to rank of dependent variable. The sorts of things we
  expect to find with  $xs\_rank = 1$  are: cross sections (to be evaluated) and rates for electron
  impact processes. In the former case, the independent variable is an energy, temperature for
  the latter. */
  if ( $xs\_rank_j \equiv 1$ ) then
    assert( $xs\_units_{1, j} \equiv$  'eV'  $\vee$   $xs\_units_{1, j} \equiv$  'eV/amu')

```

```

if (xs_eval_name_j == 'table_calc') then
  assert(xs_var_1, j == 'temperature')
  temperature = ind_var_1    // Calculate integral
  norm_constant = infinite_integral(electron_norm_integrand)
  num_evaluations = 0
  zsigma = infinite_integral(electron_rate_integrand)
  zsigma = zsigma / norm_constant
else    // Evaluate fit
  if (xs_var_1, j == 'energy' ∨ xs_var_1, j == 'temperature') then
    zte_aladdin = dble(ind_var_1)    // It expects double
  else if (xs_var_1, j == 'specific_energy') then
    assert(ra_mass_units == 'g')
    zte_aladdin = dble(ind_var_1 * ra_mass / atomic_mass_unit_g)
  else
    assert('Unexpected_value_of_xs_var[1,j]' == '_')
  end if
  zdpsigma = zero    // Doesn't initialize
  call aladdin_eval_fit(leva, zte_aladdin, fit_coef, num_fit_coef, zdpsigma, error)
  zsigma = zdpsigma
end if

/* Alternatively, treat xs_rank = 2: expect an ion impact process. */
else if (xs_rank_j == 2) then
@#if 0
  // Move this below to make specific for table_calc only
  assert(xs_units_1, j == 'eV' ∨ xs_units_1, j == 'eV/amu')
  assert(xs_units_2, j == 'eV' ∨ xs_units_2, j == 'eV/amu')
  assert(ra_mass_units == 'g')
  if ((xs_var_1, j == 'specific_energy') ∧ (xs_var_2, j == 'specific_temperature')) then
    zenergy = ind_var_1
    ztemperature = ind_var_2
  else if ((xs_var_1, j == 'specific_temperature') ∧ (xs_var_2, j == 'specific_energy')) then
    ztemperature = ind_var_1
    zenergy = ind_var_2
  else
    assert('Unexpected_values_of_xs_var' == '_')
  end if
@#endif
if (xs_eval_name_j == 'table_calc') then
  assert(xs_units_1, j == 'eV' ∨ xs_units_1, j == 'eV/amu')
  assert(xs_units_2, j == 'eV' ∨ xs_units_2, j == 'eV/amu')
  assert(ra_mass_units == 'g')
  if ((xs_var_1, j == 'specific_energy') ∧ (xs_var_2, j == 'specific_temperature')) then
    zenergy = ind_var_1
    ztemperature = ind_var_2
  else if ((xs_var_1, j == 'specific_temperature') ∧ (xs_var_2, j == 'specific_energy')) then
    ztemperature = ind_var_1
    zenergy = ind_var_2
  else
    assert('Unexpected_values_of_xs_var' == '_')
  end if
  up = sqrt(two * ztemperature)

```

```

    u = sqrt(two * zenergy)
    num_evaluations = 0
    zsigma = infinite_integral(ion_rate_integrand)
  else
    @#if 0
      // Try to generalize here, but add a specific check for JANRD where we know what the order
      // of the arguments should be.
      zte_aladdin = dble(ztemperature)
    @#endif
    assert(¬((leval ≡ '#JANRD') ∧ (xs_var1, j ≡ 'specific_energy') ∧ (xs_var2, j ≡
      'specific_temperature'))))
    if (xs_var1, j ≡ 'specific_temperature' ∨ xs_var1, j ≡ 'specific_energy') then
      assert(ra_mass_units ≡ 'g')
      zte_aladdin = dble(ind_var1 * ra_mass / atomic_mass_unit_g)
    else
      zte_aladdin = dble(ind_var1)
    end if
    num_fit_coef_aladdin = num_fit_coef + 1
    @#if 0
      fit_coef_num_fit_coef_aladdin = dble(zenergy)
    @#endif
    if (xs_var2, j ≡ 'specific_temperature' ∨ xs_var2, j ≡ 'specific_energy') then
      assert(ra_mass_units ≡ 'g')
      fit_coef_num_fit_coef_aladdin = dble(ind_var2 * ra_mass / atomic_mass_unit_g)
    else
      fit_coef_num_fit_coef_aladdin = dble(ind_var2)
    end if
    zdpsigma = zero
    call aladdin_eval_fit(leval, zte_aladdin, fit_coef, num_fit_coef_aladdin, zdpsigma, error)
    zsigma = areal(zdpsigma)
  end if
else
  assert('Ranks_higher_than_2_not_treated' ≡ ' ')
end if

table_value = zsigma

return
end

```

Compute Dependent Variable “delta”s. These steps will be executed frequently throughout the code. Perhaps this routine can be standardized and used in many places. The minimum *min\_value* and maximum *max\_value* are input. A uniform series of length *number\_of\_values*, either linear or logarithmic according to *spacing*, is to be established with the spacing *var\_delta*. At the same time, the original values are to be rescaled by *multiplier*; typically, this is used to effect a change in units. Hence, the output minimum *var\_min* may differ in general from the input *min\_value*.

⟨Functions and subroutines 2⟩ +≡

```
subroutine find_delta(spacing, min_value, multiplier, max_value, number_of_values, var_min,
    var_delta)
    implicit_none_f77
    implicit_none_f90
    real min_value, multiplier, max_value    // Input
    integer number_of_values
    character*xs_tag_string_length spacing
    real var_min, var_delta    // Output

    if (number_of_values ≤ 1) then
        var_min = zero
        var_delta = zero
        return
    end if

    if (spacing ≡ 'linear') then
        var_min = min_value * multiplier
        var_delta = (max_value - min_value) * multiplier / areal(number_of_values - 1)
    else if (spacing ≡ 'log') then
        var_min = log(min_value * multiplier)
        var_delta = log(max_value / min_value) / areal(number_of_values - 1)
    else
        assert('Unknown_spacing_type' ≡ ' ')
    end if

    return
end
```

## 4 Integrands for Rate Calculations

This routine is a single argument function which provides the integrand in a calculation of velocity-weighted averages of a user-specified function over a user-specified distribution function. The principle use of this routine is to compute a velocity-weighted reaction rate for electrons colliding with neutral particles,

$$\langle \sigma v v^n \rangle = \frac{4\pi}{\alpha m} \left( \frac{2}{m} \right)^{(n+2)/2} \int_0^\infty dE \sigma f(E) E^{(n+2)/2}, \quad (4)$$

where  $\sigma(E)$  is computed using the function *sigma*,  $f(E)$  is calculated by the function *e\_distrib\_fn*,  $m$  is the particle mass relating  $E$  and  $v$  via  $E = \frac{1}{2}mv^2$ , and  $\alpha$  is the normalization constant for the distribution function,

$$\alpha = \frac{4\pi}{m} \left( \frac{2}{m} \right)^{1/2} \int_0^\infty dE f(E) E^{1/2}.$$

Note that  $\alpha$  does not appear in this routine but is incorporated in the calling routines.

This routine is set up with a single argument so that it can be used with “canned” integration routines; the argument *energy* is the energy  $E$  in the above expressions. In order for the numerical integrations to work effectively, it is necessary that  $E$  be in practical units; eV in this case. Appropriate unit conversions have been inserted into the overall multiplier to put the result in cgs units. The mass  $m = ra\_mass$  and exponent  $n = moment\_number$  must come in through common blocks.

⟨ Functions and subroutines 2 ⟩ +=

```
function electron_rate_integrand(energy)
  implicit_none_f77
  ra_common    // Common
  implicit_none_f90
  real electron_rate_integrand
  real energy    // Input

  real multiplier, ze_dist, exponent, zint, zsigma    // Local variables
  real e_distrib_fn, sigma    // External functions

  ra_localcommon    // Local common

  zsigma = sigma(energy)
  ze_dist = e_distrib_fn(energy)
  exponent = half * (areal(moment_number) + two)
  multiplier = (const(4.0) * PI / ra_mass) * (two / ra_mass)exponent * ev_to_ergsexponent+one

  zint = multiplier * zsigma * ze_dist * energyexponent

  electron_rate_integrand = zint

return
end
```

This routine is the integrand used to compute the normalization constant for electron reaction rates. The primary motivation for doing this numerically is allow the use of arbitrary electron distribution functions. The argument *energy* is assumed to be in eV; the overall multiplier converts the resulting units to cgs.

⟨ Functions and subroutines 2 ⟩ +≡

```
function electron_norm_integrand(energy)
  implicit_none_f77
  ra_common     // Common
  implicit_none_f90
  real electron_norm_integrand
  real energy     // Input

  real multiplier, ze_dist, zint     // Local variables
  real e_distrib_fn     // External functions

  ze_dist = e_distrib_fn(energy)
  multiplier = (const(4.0) * PI / ra_mass) * sqrt(two / ra_mass) * ev_to_ergsconst(1.5)
  zint = multiplier * ze_dist * sqrt(energy)

  electron_norm_integrand = zint

return
end
```

This routine is a single argument function which provides the integrand in a calculation velocity-weighted reaction rates for cases involving two “heavy” particles. The plasma species is assumed to be described by a Maxwellian distribution function. The desired integral can be expressed in terms of [1]

$$I_{\ell,n} \equiv \frac{1}{\sqrt{\pi} u u_p^{n+1}} \int_0^\infty dv_{rel} v_{rel}^{2+n} \sigma_\ell(v_{rel}) \left\{ \exp \left[ \frac{-(v_{rel} - u)^2}{u_p^2} \right] - (-1)^n \exp \left[ \frac{-(v_{rel} + u)^2}{u_p^2} \right] \right\}, \quad (5)$$

where  $u$  is the velocity of the neutral particle, and  $u_p$  is the thermal velocity of the plasma particle. The latter is related to the plasma temperature by  $T_p = \frac{1}{2} m_p u_p^2$ . The result provided by this routine is  $I_{\ell,n} u_p^n$  and, thus, has the same dimensions as  $\langle \sigma v v^n \rangle$ .

This routine is set up with a single argument so that it can be used with “canned” integration routines; the argument *rel\_velocity* is the relative velocity  $v_{rel}$  in the above expressions. The factors  $u$  and  $u_p$ , as well as the exponent  $n$  is represented by the variable *moment\_number*. Also in common is the mass *ra\_mass* used to convert the relative velocity into an energy for use in evaluating the cross section *sigma*. The value to be used for *ra\_mass* may vary, depending on the origin of the data used in *sigma*.

The units of the various velocities are taken to be  $\sqrt{\text{eV}/\text{amu}}$  so that values of order unity for *rel\_velocity* are reasonable. It is assumed that *ra\_mass* has units of grams, that the argument of *sigma* should be in eV, and that *sigma* itself is in  $\text{cm}^2$ . The factor *multiplier* is defined so that the resulting integral has cgs units.

⟨ Functions and subroutines 2 ⟩ +≡

```
function ion_rate_integrand(rel_velocity)
  implicit_none_f77
  ra_common      // Common
  implicit_none_f90
  real ion_rate_integrand
  real rel_velocity    // Input

  real zbracket, multiplier, ze, exponent_minus,    /* Local */
      exponent_plus, zsigma
  real sigma      // External function

  ra_localcommon    // Local common

  ze = half * (ra_mass / atomic_mass_unit_g) * rel_velocity2
  zsigma = sigma(ze)
  exponent_plus = -(rel_velocity + u)2 / up2
  exponent_minus = -(rel_velocity - u)2 / up2
  zbracket = exp(exponent_minus) - (-1)moment_number * exp(exponent_plus)
  multiplier = one / (sqrt(PI) * u * up) * (sqrt(ev_to_ergs / atomic_mass_unit_g))1+moment_number

  ion_rate_integrand = multiplier * rel_velocity2+moment_number * zsigma * zbracket

return
end
```

This routine provides the electron distribution function at the energy *energy*. Presently, the distribution is assumed to be Maxwellian; the temperature *temperature* characterizing the distribution is passed in through common. Note that *temperature* and *energy* must have the same units. The routine utilizes a single argument for generality since other distributions may require additional parameters (beyond temperature) to describe them.

⟨ Functions and subroutines 2 ⟩ +≡

```
function e_distrib_fn(energy)
  implicit_none_f77
  implicit_none_f90
  real e_distrib_fn
  real energy     // Input
  ra_localcommon     // Local common
  e_distrib_fn = exp( $-energy / temperature$ )
  return
end
```

This routine serves as the intermediary between the integration functions and calls to ALADDIN subroutines. In order to fit in with the former, this function must have only a single argument; consequently, it can easily be altered without having to change the calling routines.

As presently used, this routine computes the cross section at the input energy *energy*. The other information required to communicate with ALADDIN is brought in by the common block *ra\_localcommon*.

⟨ Functions and subroutines 2 ⟩ +≡

```
function sigma(energy)
  implicit_none_f77
  implicit_none_f90
  real sigma
  real energy    // Input

  double precision ze, zfit    // Local
  character*msg_length errmsg

  ra_localcommon    // Local common

  ze = dble(energy)    // Explicitly transfer to double precision
  errmsg = '␣'
  num_evaluations = num_evaluations + 1
  call aladdin_eval_fit(leval, ze, fit_coef, num_fit_coef, zfit, errmsg)

  /* Trap errors from ALADDIN routines. Barring actual foul-ups, the the only likely error is from
     requesting cross section values at energies below threshold. A handful of the ALADDIN routines
     report this as an error condition. Assume this is the case here. */
  if (errmsg ≠ '␣') then
    zfit = 0.0 · 100D

    /* write(stderr,*)' Warning from sigma and aladdin: ',errmsg */
  end if

  sigma = areal(zfit)    // Explicitly convert back to normal precision

  return
end
```

## 5 ALADDIN Related Routines

This one reads fit coefficients. It takes as input *filename*, the name of an ALADDIN data file; *hier\_label*, an array of *num\_hier* hierarchical search labels; *bool\_and\_label*, an array of *num\_bool\_and* “and.” boolean search labels; and *bool\_not\_label*, an array of *num\_bool\_not* “not.” boolean search labels (see ALADDIN documentation for further details). Following the search through *filename*, the fit *num\_fit\_coef* fit coefficients are output in the array *cf*. The boolean label identifying the evaluation routine is returned in *leval*.

⟨ Functions and subroutines 2 ⟩ +≡

```
subroutine aladdin_read_coefs(filename, hier_label, num_hier, bool_and_label, num_bool_and,
    bool_not_label, num_bool_not, fit_coef, num_fit_coef, leval)

    implicit_none_f77
    implicit_none_f90

    character*ra_data_filename_length filename    // Input
    character*ra_search_label_length hier_label(ra_label_max), bool_and_label(ra_label_max),
        bool_not_label(ra_label_max)
    integer num_hier, num_bool_and, num_bool_not

    character*xs_eval_name_length leval    // Output
    integer num_fit_coef
    double precision fit_coef(ra_fit_coef_max)

    decl_alpcom    // Need bl and nbl from here

    logical end_of_file, match    // Local variables
    character*msg_length errmsg
    integer ientry, alad_diskin

    alad_diskin = disk_in + 1
    open(alad_diskin, file = filename, status = 'old')

    /* Loop through entries in filename */
    ientry = 0
loop: continue
    call alread(alad_diskin, ientry, end_of_file, errmsg)

    assert(¬end_of_file)
    assert(errmsg ≡ '␣')

    /* Compare this entry with labels provided on input */
    call alcomp(hier_label, num_hier, bool_and_label, num_bool_and, bool_not_label, num_bool_not, match)
    if (¬match)
        goto loop

    close(unit = alad_diskin)    // Have a match

    /* Decode coefficients; check first that access label is the standard one */
    assert(bl1SP1:2 ≡ '$␣')

    call alrecf(fit_coef, num_fit_coef, ra_fit_coef_max, errmsg)
    assert(errmsg ≡ '␣')

    leval = blnbl    // Assign evaluation label variable

    /* Note: this step is not really necessary; could just return bl and select the last value when
        calling the evaluation routine. */
    return

end
```

Evaluate ALADDIN fits. This routine does the dirty job of associating the input boolean label for the evaluation routine, *leval*, with an actual FORTRAN subroutine. Because all of the ALADDIN routines of interest currently have essentially the same arguments (although they differ in details which are of no consequence to the compiler), this clearing-house routine can be generated automatically from a shell script which reads through the file containing all of the ALADDIN subroutines.

In addition to *leval*, the input arguments are: *pet*, the energy or temperature at which the ALADDIN fit is to be evaluated and *fit\_coef*, the array of *num\_fit\_coef* fit coefficients passed in from the ALADDIN data files. The resulting value is returned as *pfit*. Errors are returned via *kermsg*.

⟨Functions and subroutines 2⟩ +=

```
subroutine aladdin_eval_fit(leval, pet, fit_coef, num_fit_coef, pfit, kermsg) implicit_none_f77
implicit_none_f90
```

```
character*xs_eval_name_length leval     // Input
```

```
integer num_fit_coef
```

```
double precision pet
```

```
double precision fit_coef(ra_fit_coef_max)
```

```
double precision pfit     // Output
```

```
character*msg_length kermsg
```

```
if (leval ≡ '#ALMEWE') then
```

```
    call ALMEWE(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#BELI') then
```

```
    call BELI(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#ALCHEB') then
```

```
    call ALCHEB(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#ALKING') then
```

```
    call ALKING(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#ALPHACX') then
```

```
    call ALPHACX(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#JAN1') then
```

```
    call JAN1(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#ETABSQ') then
```

```
    call ETABSQ(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#FNAGEX') then
```

```
    call FNAGEX(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#FNRC') then
```

```
    call FNRC(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#FNXS') then
```

```
    call FNXS(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#FORC') then
```

```
    call FORC(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#FOXS') then
```

```
    call FOXS(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#JBORN1') then
```

```
    call JBORN1(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#JBORN2') then
```

```
    call JBORN2(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#JANRD') then
```

```
    call JANRD(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#EEXCH2') then
```

```
    call EEXCH2(pet, fit_coef, num_fit_coef, pfit, kermsg)
```

```
else if (leval ≡ '#EIONH2') then
```

```

    call EIONH2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#EIONHR') then
    call EIONHR(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JRREC1') then
    call JRREC1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JRREC3') then
    call JRREC3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC1') then
    call JEEXC1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC2') then
    call JEEXC2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC3') then
    call JEEXC3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC4') then
    call JEEXC4(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC6') then
    call JEEXC6(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JIONHE') then
    call JIONHE(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEION5') then
    call JEION5(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JRREC2') then
    call JRREC2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC5') then
    call JEEXC5(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JEEXC7') then
    call JEEXC7(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JDREC1') then
    call JDREC1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#ALJAN3') then
    call ALJAN3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#PEXCH4') then
    call PEXCH4(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#IONBEA') then
    call IONBEA(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JCX1') then
    call JCX1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#JCX2') then
    call JCX2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#NEEXH1') then
    call NEEXH1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#NEEXH2') then
    call NEEXH2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#EEXCJN') then
    call EEXCJN(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#EIONJN') then
    call EIONJN(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXC1') then
    call HEXC1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXC2') then
    call HEXC2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXC3') then

```

```

call HEXC3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXC4') then
    call HEXC4(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXC5') then
    call HEXC5(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HEXCLD') then
    call HEXCLD(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HIONN') then
    call HIONN(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HCX1') then
    call HCX1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HCX2') then
    call HCX2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HCX3') then
    call HCX3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#AQEXC1') then
    call AQEXC1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#AQEXC2') then
    call AQEXC2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#AQEXC3') then
    call AQEXC3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#AIONGL') then
    call AIONGL(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#HCX4') then
    call HCX4(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#ACXGL1') then
    call ACXGL1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#ACXGL2') then
    call ACXGL2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#REFL1') then
    call REFL1(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#REFL2') then
    call REFL2(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#REFL3') then
    call REFL3(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#SPTEK') then
    call SPTEK(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#SPTTH') then
    call SPTTH(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#ELCROSS') then
    call ELCROSS(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#ELSPLINE') then
    call ELSPLINE(pet, fit_coef, num_fit_coef, pfit, kermmsg)
else if (leval == '#INTERP2D') then /* INTERP2D, like other Aladdin routines explicitly uses
double precision. However, for convenience it makes use of the find_delta and interpolate2
routines. This assert will alert the user to a precision incompatibility with those routines. */
assert ( index ( $STRING ( real ), 'DOUBLE' ) > 0 )
call INTERP2D(pet, fit_coef, num_fit_coef, pfit, kermmsg)
end if

return end

```

## 6 Integration Routines

This first function is used to hide some of the machinery needed to compute the infinite integrals required by this code. It is assumed that all integrals have limits  $0 \rightarrow \infty$ . In practice, however, the integrand usually exhibits some sort of threshold behavior so that the actual lower limit is nonzero. The integrands typically vanish exponentially for large values of the variable of integration. Hence, there is also a practical upper limit to the integral where the integrand vanishes numerically. In addition, the integrand may utilize a fit function of some sort and take a zero value outside the range of validity of that fit. The upshot is again that the range of integration is effectively smaller than  $0 \rightarrow \infty$ . Since these discontinuities, particularly the last type, can cause undue strain on the integration routine, it is worthwhile to explicitly locate these “thresholds” and modify the limits of integration accordingly. Additional function evaluations are used, perhaps more than the integrator would otherwise need, but the result is a more robust procedure.

Even with a finite range of integration, a variable transformation may be used to shrink the range to fit within the interval  $0 \rightarrow 1$ . For the transformation used here, this is difficult to do with a single transformation if the lower limit is still  $< 1$ . Consequently, the interval may be split into two parts so that separate transformations can be used on each.

**Note:** The choice of *dcadre* as the integration routine used here represents a compromise. On the one hand, the routine needs to be portable and easy to distribute (e.g., this requirement rules out most commercial packages). On the other hand, the method must be robust and relatively efficient. Because discontinuities in the sorts of integrands to be used here are not unusual (particularly in the first derivative), simple high order schemes fail miserably. While a trapezoidal rule works admirably in these cases, it is terribly slow, requiring many millions of function evaluations to achieve a reasonable accuracy. A proper treatment of these situations requires an adaptive routine; the literature indicates that an efficient implementation of an adaptive scheme is tricky. This led to the identification of *dcadre* as a candidate routine, a public-domain routine that has been used previously in commercial libraries.

Even then, it was decided that the sudden onsets at thresholds and fit limits was causing *dcadre* to struggle excessively. For this reason, the *find\_threshold* procedure was implemented. The use of a variable transformation in combination with finite limits appeared to further improve the efficiency of the integrations.

If an integral fails to converge, the user can try the following:

1. Increase the error tolerance, *integration\_acc*.
2. Try turning off the variable transformation; i.e., set *integ\_transform\_ratio* = 0.
3. Eliminate the threshold calculation (replace bounds in calls to *dcadre* with the original ones).
4. Get a better integration routine! (Better ones may exist).

⟨ Functions and subroutines 2 ⟩ +≡

```

function infinite_integral(integrand)
  implicit_none_f77
  implicit_none_f90
  real infinite_integral
  integer ier    // Local
  real dcadre, lower_result, upper_result, error, lower_bound_rev, upper_bound_rev
  external integrand    // External
  real integrand

  /* Locate “thresholds” if any and alter bounds accordingly; the accuracy used here is the same as
     that used for the integrator. */

  call find_threshold(zero, integration_infinity, integrand, integration_acc, lower_bound_rev,
    upper_bound_rev)

  /* If lower limit is still < 1 (e.g., possibly still zero), split integral into two pieces so that a
     variable transformation can be used on the upper half if needed. */
  if (lower_bound_rev < one) then
    lower_result = dcadre(integrand, lower_bound_rev, one, zero, integration_acc, error, ier)
    assert(ier < 100)
    upper_result = dcadre(integrand, one, upper_bound_rev, zero, integration_acc, error, ier)
    assert(ier < 100)
  else
    lower_result = zero
    upper_result = dcadre(integrand, lower_bound_rev, upper_bound_rev, zero, integration_acc,
      error, ier)
    assert(ier < 100)
  end if

  infinite_integral = lower_result + upper_result

  return
end

```

Cautious adaptive Romberg extrapolation integrator. This routine computes  $\int_a^b f(x)dx$  using a cautious adaptive Romberg extrapolation. The user provides  $f$ , a single-argument real function subprogram; it must be declared external in the calling program.  $aerr$  is the desired *absolute* error in the answer;  $rerr$  is the desired *relative* error in the answer. On output  $dcadre$  returns  $error$ , an estimated bound on the absolute error of the integral. Non-zero values of the error parameter  $ier$  indicate one of the following conditions:

- Warning error with  $ier = 64 + n$ ,
  - $n = 1$  implies that one or more singularities were successfully handled.
  - $n = 2$  implies that, in some subinterval(s), the estimate of the integral was accepted merely because the estimated error was small, even though no regular behavior was recognized.
- Terminal error with  $ier = 128 + n$ ,
  - $n = 3$ , failure due to insufficient internal working storage.
  - $n = 4$ , failure; may be due to too much noise in the function (relative to the given error requirements) or due to an ill-behaved integrand.
  - $n = 5$  indicates that  $rerr > 0.1$ ,  $rerr < 0$ , or  $rerr$  is too small for the precision of the machine.

This routine is similar to one found in commercial numerical software packages. References [4] and [5] describe this routine and provide text versions of the source code. It is assumed that this routine is in the public domain. The code has been altered slightly to make it compatible with the rest of this (WEB) program. Otherwise, the “latest revision” date of January 27, 1977 is still applicable.

⟨ Functions and subroutines 2 ⟩ +≡

```

function dcadre(f, lower_bound, upper_bound, aerr, rerr, error, ier)
  implicit_none_f77
  implicit_none_f90
  real dcadre
  real lower_bound, upper_bound, f, aerr, rerr    // Input
  integer ier    // Output
  real error

  integer i, ibeg, iend, ii, iii, istage, istep, istep2,    /* Local */
    it, l, lm1, maxtbl, maxts, mxtstge, n, n2, nnleft
  real a, absi, aitlow, aittol, alg4o2, alpha, astep, b, beg, cadre, curest, diff, edn, ergrl, ergoal, erra,
    error, erret, errr, f_sub, fbeg, fbeg2, fend, fextm1, fextrp, fi, fn, fnsize, four, fourp5, h2next,
    h2tfex, h2tol, hovn, hun, jump1, length, p1, prever, sing, singnx, slope, stage, step, stepmn,
    stepnm, sum, sumabs, tabs, tabt1m, ten, x, vint
  real t10, 10, r10, ait10, dif10, rn4, ts2049, ibegs30, begin30, finis30, est30
  logical h2conv, inf_integ, aitken, right, reglar
  logical reglsv30

  data maxts, maxtbl, mxtstge / 2049, 10, 30 /

  f_sub(x) = f(x-integ-transform-exp) * (areal(integ-transform-exp) / xinteg-transform-exp+1)

  /* In the original version the following variables were assigned in data statements. We use the
     “const” macro here to ensure that proper precision is used. */
  aitlow = const(1.1)
  h2tol = const(0.15)
  aittol = const(0.1)
  jump1 = const(0.01)
  rn1 = const(0.7142005)
  rn2 = const(0.3466282)
  rn3 = const(0.843751)
  rn4 = const(0.1263305)
  p1 = const(0.1)
  four = const(4.0)
  fourp5 = const(4.5)
  ten = const(10.0)
  hun = const(100.0)

  alg4o2 = log10(two)    // Initializations
  cadre = zero
  error = zero
  curest = zero
  vint = zero
  ier = 0

  if (max(one, lower_bound) / upper_bound < integ-transform-ratio) then
    a = upper_bound-one / areal(integ-transform-exp)
    b = lower_bound-one / areal(integ-transform-exp)
    inf_integ = T

```

```

    else
        a = lower_bound
        b = upper_bound
        inf_integ =  $\mathcal{F}$ 
    end if

    length = abs(b - a)
    if (length  $\equiv$  zero)
        go to 215
    if (rerr > p1  $\vee$  rerr < zero)
        go to 210
    if (aerr  $\equiv$  zero  $\wedge$  (rerr + hun)  $\leq$  hun)
        go to 210
    errr = rerr
    erra = abs(aerr)
    stepmn = (length / float( $2^{m_{xstge}}$ ))
    stepnm = max(length, abs(a), abs(b)) * ten
    stage = half
    istage = 1
    fsize = zero
    prever = zero
    reglar =  $\mathcal{F}$ 

    /* The given interval of integration is the first interval considered. */
    beg = a
    if (inf_integ) then
        fbeg = f_sub(beg) * half
    else
        fbeg = f(beg) * half
    end if
    ts1 = fbeg
    ibeg = 1
    edn = b
    if (inf_integ) then
        fend = f_sub(edn) * half
    else
        fend = f(edn) * half
    end if
    ts2 = fend
    iend = 2
5: right =  $\mathcal{F}$ 

    /* Investigation of a particular subinterval begins at this point. */
10: step = edn - beg
    astep = abs(step)
    if (astep < stepmn)
        go to 205
    if (stepnm + astep  $\equiv$  stepnm)
        go to 205
    t1,1 = fbeg + fend
    tabs = abs(fbeg) + abs(fend)
    l = 1
    n = 1
    h2conv =  $\mathcal{F}$ 

```

```

    aitken =  $\mathcal{F}$ 
15: lm1 = l
    l = l + 1

    /* Calculate the next trapezoid sum,  $t_{l, 1}$ , which is based on  $n2 + 1$  equispaced points. Here,
        $n2 = 2n = 2^{l-1}$ . */
    n2 = n + n
    fn = n2
    istep = (iend - ibeg) / n
    if (istep > 1)
        go to 25
    ii = iend
    iend = iend + n
    if (iend > maxts)
        go to 200
    hovn = step / fn
    iii = iend
    fi = one
    do i = 1, n2, 2
        tsiii = tsii
        if (inf_integ) then
            tsiii-1 = f_sub(edn - fi * hovn)
        else
            tsiii-1 = f(edn - fi * hovn)
        end if
        fi = fi + two
        iii = iii - 2
        ii = ii - 1
    end do
    istep = 2
25: istep2 = ibeg + istep / 2
    sum = zero
    sumabs = zero
    do i = istep2, iend, istep
        sum = sum + tsi
        sumabs = sumabs + abs(tsi)
    end do
     $t_{l, 1} = t_{l-1, 1} * half + sum / fn$ 
     $tabs = tabs * half + sumabs / fn$ 
    absi = astep * tabs
    n = n2

    /* Get preliminary value for vint from last trapezoid sum and update the error requirement ergoal
       for this subinterval. */
    it = 1
    vint = step *  $t_{l, 1}$ 
    tabtln = tabs * ten
    fnsiz = max(fnsiz, abs( $t_{l, 1}$ ))
    ergl = astep * fnsiz * ten
    ergoal = stage * max(erra, errr * abs(curest + vint))

    /* Complete row l and column l of t array. */
    fextrp = one
    do i = 1, lm1

```

```

    fextrp = fextrp * four
     $t_{i, l} = t_{l, i} - t_{l-1, i}$ 
     $t_{l, i+1} = t_{l, i} + t_{i, l} / (fextrp - one)$ 
end do
errerr = astep * abs( $t_{1, l}$ )

    /* Preliminary decision procedure: if  $l = 2$  and  $t_{2, 1} = t_{1, 1}$ , go to 135 to follow up the impression
       that integrand is a straight line. */
if ( $l > 2$ )
    go to 40
if ( $tabs + p1 * abs(t_{1, 2}) \equiv tabs$ )
    go to 135
go to 15

    /* Calculate next ratios for columns 1,...,l-2 of  $t$ -table. Ratio is set to zero if difference in last two
       entries of column is about zero. */
40: do  $i = 2, lm1$ 
    diff = zero
    if ( $tabt_{lm} + abs(t_{i-1, l}) \neq tabt_{lm}$ )
         $diff = t_{i-1, lm1} / t_{i-1, l}$ 
         $t_{i-1, lm1} = diff$ 
    end do
if ( $abs(four - t_{1, lm1}) \leq h2tol$ )
    go to 60
if ( $t_{1, lm1} \equiv zero$ )
    go to 55
if ( $abs(two - abs(t_{1, lm1})) < jump_{tl}$ )
    go to 130
if ( $l \equiv 3$ )
    go to 15
h2conv =  $\mathcal{F}$ 
if ( $abs((t_{1, lm1} - t_{1, l-2}) / t_{1, lm1}) \leq aittol$ )
    go to 75
50: if (reglar)
    go to 55
if ( $l \equiv 4$ )
    go to 15
55: if ( $error > ergoal \wedge (ergl + error) \neq ergl$ )
    go to 175
    go to 145     /* Cautious Romberg extrapolation */
60: if (h2conv)
    go to 65
    aitken =  $\mathcal{F}$ 
    h2conv =  $\mathcal{T}$ 
65: fextrp = four
70:  $it = it + 1$ 
     $vint = step * t_{l, it}$ 
     $error = abs(step / (fextrp - one) * t_{it-1, l})$ 
    if ( $error \leq ergoal$ )
        go to 160
    if ( $ergl + error \equiv ergl$ )
        go to 160
    if ( $it \equiv lm1$ )

```

```

    go to 125
  if ( $t_{it, lm1} \equiv zero$ )
    go to 70
  if ( $t_{it, lm1} \leq fextrp$ )
    go to 125
  if ( $abs(t_{it, lm1} / four - fextrp) / fextrp < aittol$ )
     $fextrp = fextrp * four$ 
  go to 70

  /* Integrand may have  $x^\alpha$  type singularity resulting in a ratio of  $sing = 2^{\alpha+1}$ . */
75: if ( $t_{1, lm1} < aittlow$ )
  go to 175
  if ( $aitken$ )
    go to 80
   $h2conv = \mathcal{F}$ 
   $aitken = \mathcal{T}$ 
80:  $fextrp = t_{l-2, lm1}$ 
  if ( $fextrp > fourp5$ )
    go to 65
  if ( $fextrp < aittlow$ )
    go to 175
  if ( $abs(fextrp - t_{l-3, lm1}) / t_{1, lm1} > h2tol$ )
    go to 175
   $sing = fextrp$ 
   $fextm1 = one / (fextrp - one)$ 
   $ait_1 = zero$ 
  do  $i = 2, l$ 
     $ait_i = t_{i, 1} + (t_{i, 1} - t_{i-1, 1}) * fextm1$ 
     $r_i = t_{1, i-1}$ 
     $dif_i = ait_i - ait_{i-1}$ 
  end do
   $it = 2$ 
90:  $vint = step * ait_l$ 
   $error = error * fextm1$ 
  if ( $error > ergoal \wedge (ergl + error) \neq ergl$ )
    go to 95
   $alpha = \log_{10}(sing) / alg4o2 - one$ 
   $ier = \max(ier, 65)$ 
  go to 160
95:  $it = it + 1$ 
  if ( $it \equiv lm1$ )
    go to 125
  if ( $it > 3$ )
    go to 100
   $h2next = four$ 
   $singnx = sing + sing$ 
100: if ( $h2next < singnx$ )
  go to 105
   $fextrp = singnx$ 
   $singnx = singnx + singnx$ 
  go to 110
105:  $fextrp = h2next$ 
   $h2next = four * h2next$ 

```

```

110: do  $i = it, lm1$ 
       $r_{i+1} = zero$ 
      if ( $tabtln + abs(dif_{i+1}) \neq tabtln$ )
         $r_{i+1} = dif_i / dif_{i+1}$ 
      end do
       $h2tfex = -h2tol * fextrp$ 
      if ( $r_l - fextrp < h2tfex$ )
        go to 125
      if ( $r_{l-1} - fextrp < h2tfex$ )
        go to 125
       $error = astep * abs(dif_l)$ 
       $fextm1 = one / (fextrp - one)$ 
      do  $i = it, l$ 
         $ait_i = ait_i + dif_i * fextm1$ 
         $dif_i = ait_i - ait_{i-1}$ 
      end do
      go to 90

      /* Current trapezoid sum and resulting extrapolated values did not give a small enough error.
      Note, having  $prever < error$  is an almost certain sign of beginning trouble with in the function
      values. Hence, a watch for, and control of, noise should begin here. */
125:  $fextrp = \max(prever / error, aitlow)$ 
       $prever = error$ 
      if ( $l < 5$ )
        go to 15
      if ( $l - it > 2 \wedge istage < mxstage$ )
        go to 170
       $erret = error / (fextrp^{maxtbl-l})$ 
      if ( $erret > ergoal \wedge (ergl + erret) \neq ergl$ )
        go to 170
      go to 15

      /* Integrand has jump */
130: if ( $error > ergoal \wedge (ergl + error) \neq ergl$ )
      go to 170

       $diff = abs(t_{1,l}) * (fn + fn)$      // Note that  $2fn = 2^l$ 
      go to 160

      /* The integrand is a straight line: test this assumption by comparing the value of the integrand
      at four randomly chosen points with the value of the straight line interpolating the integrand at
      the two end points of the sub-interval. If test is passed, accept  $vint$ . */
135:  $slope = (fend - fbeg) * two$ 
       $fbeg2 = fbeg + fbeg$ 
      do  $i = 1, 4$ 
        if ( $inf\_integ$ ) then
           $diff = abs(f\_sub(beg + rn_i * step) - fbeg2 - rn_i * slope)$ 
        else
           $diff = abs(f(beg + rn_i * step) - fbeg2 - rn_i * slope)$ 
        end if
        if ( $tabtln + diff \neq tabtln$ )
          go to 155
      end do
      go to 160

```

```

    /* Noise may be a dominant feature: estimate noise level by comparing the value of the integrand
       at four randomly chosen points with the value of the straight line interpolating the integrand at
       the two endpoints. If small enough, accept vint. */
145: slope = (fend - fbeg) * two
    fbeg2 = fbeg + fbeg
    i = 1
150: if (inf_integ) then
    diff = abs(f_sub(beg + rni * step) - fbeg2 - rni * slope)
    else
    diff = abs(f(beg + rni * step) - fbeg2 - rni * slope)
    end if
155: error = max(error, astep * diff)
    if (error > ergoal ∧ (ergl + error) ≠ ergl)
    go to 175
    i = i + 1
    if (i ≤ 4)
    go to 150
    ier = 66

    /* Intergration over current subinterval successful: add vint to dcadre and error to error, then
       set up next subinterval, if any. */
160: cadre = cadre + vint
    error = error + error
    if (right)
    go to 165
    istage = istage - 1
    if (istage ≡ 0)
    go to 220
    reglar = reglsvistage
    beg = beginistage
    edn = finisistage
    curest = curest - estistage+1 + vint
    iend = ibeg - 1
    fend = tsiend
    ibeg = ibegsistage
    go to 180
165: curest = curest + vint
    stage = stage + stage
    iend = ibeg
    ibeg = ibegsistage
    edn = beg
    beg = beginistage
    fend = fbeg
    fbeg = tsibeg
    go to 5

    /* Integration over current subinterval is unsuccessful; mark subinterval for further subdivision.
       Set up next subinterval. */
170: reglar = T
175: if (istage ≡ mxstge)
    go to 205
    if (right)
    go to 185

```

```

      reglsvistage+1 = reglar
      beginistage = beg
      ibegsistage = ibeg
      stage = stage * half
180: right = T
      beg = (beg + edn) * half
      ibeg = (ibeg + iend) / 2
      tsibeg = tsibeg * half
      fbeg = tsibeg
      go to 10
185: nnleft = ibeg - ibegsistage
      if (iend + nnleft ≥ maxts)
        go to 200
      iii = ibegsistage
      ii = iend
      do i = iii, ibeg
        ii = ii + 1
        tsii = tsi
      end do
      do i = ibeg, ii
        tsiii = tsi
        iii = iii + 1
      end do
      iend = iend + 1
      ibeg = iend - nnleft
      fend = fbeg
      fbeg = tsibeg
      finisistage = edn
      edn = beg
      beg = beginistage
      beginistage = edn
      reglsvistage = reglar
      istage = istage + 1
      reglar = reglsvistage
      estistage = vint
      curest = curest + estistage
      go to 5

      /* Failure to handle given integration problem */
200: ier = 131
      go to 215
205: ier = 132
      go to 215
210: ier = 133
215: cadre = curest + vint
220: dcadre = cadre
9000: continue
9005: return
      end

```

Locate thresholds. This routine is designed to pin down the locations of “thresholds” in the input function. The procedure is loosely related to that of finding a root, except that the function may be discontinuous at the threshold so that a very robust (bisection) method must be used. An initial interval  $lower\_bound < x < upper\_bound$  is input. The function  $f\_of\_x$  is specified externally. The routine will return revised bounds  $lower\_bound\_rev$  and  $upper\_bound\_rev$  such that  $f\_of\_x(lower\_bound\_rev) \neq 0$ , but  $f\_of\_x(lower\_bound\_rev - \epsilon) = 0$  and  $f\_of\_x(upper\_bound\_rev) \neq 0$  with  $f\_of\_x(upper\_bound\_rev + \epsilon) = 0$ , where  $\epsilon$  is the input parameter *error*.

⟨Functions and subroutines 2⟩ +≡

**subroutine** *find\_threshold*(*lower\_bound*, *upper\_bound*, *f\_of\_x*, *error*, *lower\_bound\_rev*, *upper\_bound\_rev*)

*implicit\_none\_f77*

*implicit\_none\_f90*

**real** *lower\_bound*, *upper\_bound*, *f\_of\_x*, *error*     // Input

**real** *lower\_bound\_rev*, *upper\_bound\_rev*     // Output

**real** *f\_lower*, *f\_upper*, *guess*, *lower\_guess*,     /\* Local \*/  
       *upper\_guess*, *bisect*

**external** *f\_of\_x*     // External

*f\_lower* = *f\_of\_x*(*lower\_bound*)

*f\_upper* = *f\_of\_x*(*upper\_bound*)

*lower\_bound\_rev* = *lower\_bound*     // Default values

*upper\_bound\_rev* = *upper\_bound*

/\* If *f\_of\_x* is nonzero at one of the limits, use that limit with *bisect* to find the threshold at the other end. Can then return. \*/

**if** ((*f\_lower*  $\neq$  zero)  $\vee$  (*f\_upper*  $\neq$  zero)) **then**

**if** (*f\_upper*  $\equiv$  zero)

*upper\_bound\_rev* = *bisect*(*upper\_bound*, *lower\_bound*, *error*, *f\_of\_x*)

**if** (*f\_lower*  $\equiv$  zero)

*lower\_bound\_rev* = *bisect*(*lower\_bound*, *upper\_bound*, *error*, *f\_of\_x*)

**else**

  /\* Zero at both limits. Since *find\_nonzero* cannot accept a bound of zero, replace *lower\_bound* with *integration\_acc*. But, first check to see if *f\_of\_x* is nonzero there; if so, do not need to call *find\_nonzero*! \*/

**if** (*lower\_bound*  $\equiv$  zero) **then**

**if** (*f\_of\_x*(*integration\_acc*)  $\neq$  zero) **then**

*lower\_guess* = zero

*guess* = *integration\_acc*

*upper\_guess* = *upper\_bound*

**else**

**call** *find\_nonzero*(*integration\_acc*, *upper\_bound*, *lower\_guess*, *guess*, *upper\_guess*, *f\_of\_x*)

**end if**

**else**

**call** *find\_nonzero*(*lower\_bound*, *upper\_bound*, *lower\_guess*, *guess*, *upper\_guess*, *f\_of\_x*)

**end if**

  /\* If *find\_nonzero* fails, it will set the upper and lower guesses equal. Otherwise, use *bisect* to find the two thresholds. \*/

**if** (*lower\_guess*  $\neq$  *upper\_guess*) **then**

*upper\_bound\_rev* = *bisect*(*upper\_guess*, *guess*, *error*, *f\_of\_x*)

*lower\_bound\_rev* = *bisect*(*lower\_guess*, *guess*, *error*, *f\_of\_x*)

**end if**

**end if**

```
    return  
end
```

Find a nonzero value of a function. Since nothing is assumed to be known a priori about the function  $f\_of\_x$ , this routine must be able to cover the input interval  $lower\_bound$  to  $upper\_bound$  thoroughly. For this reason, a sequence of successively finer subdivisions, much like that used to incrementally refine a trapezoidal integration, is used to break up the interval into smaller and smaller pieces. Of course, the routine is essentially done once *one* nonzero value of  $f\_of\_x$  is found. This value of  $x$  is returned as  $guess$ . One virtue of the approach used here is that adjacent  $x$  at which  $f\_of\_x = 0$  are already known from previous evaluations. These are returned as  $lower\_guess$  and  $upper\_guess$ .

Note that because a logarithmic subdivision is used, both of the bounds should be positive.

The parameter used to limit the number of subdivisions made is  $min\_bound\_ratio$ . It represents the minimum ratio of  $upper\_guess$  to  $lower\_guess$  which could be resolved by this routine. The value of  $min\_bound\_ratio$  should not be too small since the maximum number of function evaluations is  $2^{max\_stage} - 1$ ; i.e., a large number.

⟨Functions and subroutines 2⟩ +≡

```
subroutine find_nonzero(lower_bound, upper_bound, lower_guess, guess, upper_guess, f_of_x)

  implicit none
  implicit none

  real lower_bound, upper_bound, f_of_x    // Input
  real lower_guess, guess, upper_guess    // Output
  integer i, istage, max_stage    // Local
  real log_lower, log_upper, log_min_ratio, delta, log_guess

  assert(upper_bound > zero)
  assert(lower_bound > zero)    /* Divide up the interval (in log10 space). The maximum number of
    function evaluations is  $2^{max\_stage} - 1$ . */
  log_upper = log10(upper_bound)
  log_lower = log10(lower_bound)
  log_min_ratio = log10(min_bound_ratio)
  max_stage = (log((log_upper - log_lower) / log_min_ratio) / log(two)) + one

  do istage = 1, max_stage

    /* Calculate the step size (in log10 space) and log10 of the first x for each istage. */
    delta = (log_upper - log_lower) / twoistage-1
    log_guess = log_lower + half * delta

    /* Each stage consists of evaluating f_of_x at only those new values of x required to achieve a
       grand total of  $2^{istage} - 1$  evaluations. */
    do i = 1, 2istage-1
      guess = const(1., 1)log_guess
      if (f_of_x(guess) ≠ zero)
        goto loop
      log_guess = log_guess + delta
    end do
  end do

  /* Cannot find a point where f_of_x ≠ 0; indicate this condition by setting lower_guess =
     upper_guess. */
  lower_guess = lower_bound
  upper_guess = lower_guess
return

loop: continue
```

```

    /* Having found a nonzero value of f_of_x, know that the points (in log space) delta/2 above and
       below have f_of_x = 0. */
    upper_guess = const(1., 1)log_guess+half*delta
    lower_guess = const(1., 1)log_guess-half*delta

    return
end

```

Interval bisection routine. This function is called by *find\_threshold* to further bisect the input interval to localize the threshold to the specified accuracy. The first argument *zero\_side* is the side of the interval at which *f\_of\_x* = 0; likewise, at *nonzero\_side*, *f\_of\_x* ≠ 0. It is assumed that *find\_threshold* has arranged these in the proper order, and they are not checked here. The interval is bisected until it is *error* wide. The function returns the bound of that interval at which *f\_of\_x* ≠ 0.

⟨ Functions and subroutines 2 ⟩ +≡

```

function bisect(zero_side, nonzero_side, error, f_of_x)
    implicit_none_f77
    implicit_none_f90
    real bisect
    real zero_side, nonzero_side, error, f_of_x    // Input
    integer i, ibound    // Local
    real rev_zero, rev_nonzero, guess

    /* Determine number of bisections required to shrink interval to error */
    ibound = (log(abs(nonzero_side - zero_side) / error) / log(two)) + 1

    rev_zero = zero_side
    rev_nonzero = nonzero_side
    do i = 1, ibound
        guess = half * (rev_zero + rev_nonzero)
        if (f_of_x(guess) ≡ zero) then
            rev_zero = guess
        else
            rev_nonzero = guess
        end if
    end do

    bisect = rev_nonzero    // Assign point with last nonzero f_of_x

    return
end

```

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⟨ Memory allocation interface 0 ⟩ Used in section 3.

**COMMAND LINE:** "fweave -f -i! -W[ -ykw700 -ytw40000 -j -n/  
/u/dstotler/degas2/src/ratecalc.web".

**WEB FILE:** "/u/dstotler/degas2/src/ratecalc.web".

**CHANGE FILE:** (none).

**GLOBAL LANGUAGE:** FORTRAN.